Q1.

In 2007, a new law came in to stop people smoking in public buildings.

(a) Smoking can be very harmful.
Which three problems can be caused by smoking?

Tick the **three** correct boxes.

being out of breath easily	☐	lung cancer	☐
being overweight	☐	food poisoning	☐
heart disease	☐		

2 marks

(b) Some scientists investigate 'passive smoking'. Passive smoking is when people breathe in smoke from other people's cigarettes.
They checked the health of three groups of people.

group A	group B	group C
non-smokers who spend **no** time in smoky places	**non-smokers** who spend time in smoky places	**smokers** who spend time in smoky places

(i) Which group of people breathe in the **least** cigarette smoke?
Tick the correct box.

group A ☐ group B ☐ group C ☐

1 mark

(ii) Which **two** groups will help scientists find out the effects of passive smoking?
Tick the **two** correct boxes.

group A ☐ group B ☐ group C ☐

1 mark

(c) People in **group B** are likely to have similar health problems to people in **group C.**
Explain why.

.....

.....

1 mark

(d) Four scientists investigated passive smoking. The table below shows the number of people each scientist studied from each group.

scientist	group A	group B	group C
David	289	3	18
Olga	8	6	11
Peter	402	399	403
Mary	15	210	511

Which scientist is likely to get the most reliable results?
Tick the correct box.

David ☐ Olga ☐

Peter ☐ Mary ☐

1 mark
maximum 6 marks

Q1.

(a) • being out of breath easily ✓

• heart disease ✓

• lung cancer ✓

*if **all three** answers are correct, award two marks*
*if **two** answers are correct, award one mark*
*if only **one** answer is correct, award no mark*
if more than three boxes are ticked, deduct one mark for each incorrect tick

2 (L3)

(b) (i) • group A ✓

if more than one box is ticked, award no mark

1 (L3)

(ii) • group A ✓
group B ✓

***both** boxes are required for the mark*
if more than two boxes are ticked, award no mark

1 (L4)

(c) • they both breathe in smoke

accept 'they spend time in smoky places'
accept 'they breathe in second-hand smoke'
accept 'they are passive smokers'

*do **not** accept 'they both smoke'*

1 (L4)

(d) • Peter ✓

if more than one box is ticked, award no mark

1 (L4)

[6]